

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.4: Independence, bases, and dimension

© 2024–2026 Jacob Denson, Kaitlyn Phillipson, Sebastien Roch. Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Linear independence

Activity: Comparing two spanning sets (U2-LO4)

In A set described by two parameters (U2-LO4) we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In Writing a parametric set as a span (U2-LO4), we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W . The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W . But why is S ‘better’?

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the ‘most efficient’. In data terms, independence asks whether one feature direction is already built from the others.

Definition: Linear Independence

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be

linearly independent

, or simply

independent

if the only solution to the equation $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$ is trivial, i.e., $t_1 = \cdots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is

linearly dependent

if it

is not linearly independent, i.e., if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$.

Remark: An independent list cannot contain the zero vector

Any list containing $\mathbf{0}$ is linearly dependent. Give the zero vector coefficient 1 and every other vector coefficient 0:

$$1 \cdot \mathbf{0} + 0\mathbf{x}_2 + \cdots + 0\mathbf{x}_k = \mathbf{0}.$$

This linear combination equals the zero vector, but its coefficients are not all zero.

Activity: Dependence of two vectors (U2-L04)

Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero vectors in $\mathbb{R}^n$. Explain why $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel, i.e., one vector is a scalar multiple of the other.

Example: Testing independence in three dimensions (U2-LO4)

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

We need to determine the solutions to the equation $t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + t_3 \mathbf{v}_3 = \mathbf{0}$. Expanding out this

equation gives the system

$$\begin{aligned}3t_1 + t_2 - t_3 &= 0 \\2t_1 + 3t_2 + 4t_3 &= 0 \\2t_1 + 0t_2 - 2t_3 &= 0.\end{aligned}$$

Practice this system in The Elimination Scratchpad

We solve this homogeneous system of linear equations using the Gaussian elimination algorithm. First we create the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right]$$

We reduce the matrix to row echelon form, scaling each pivot row to obtain a leading one before eliminating entries below it:

$$\begin{aligned}
& \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right] \\
& \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right] \\
& \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & 0 \\ 0 & -\frac{2}{3} & -\frac{4}{3} & 0 \end{array} \right] \\
& \xrightarrow{R_2 \rightarrow \frac{3}{7}R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & -\frac{2}{3} & -\frac{4}{3} & 0 \end{array} \right] \\
& \xrightarrow{R_3 \rightarrow R_3 + \frac{2}{3}R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]
\end{aligned}$$

There is one free variable corresponding to this matrix, which implies that there is one basic solution to the equation, and thus the system has a nontrivial solution. Thus the vectors are

not linearly independent.

Activity: Testing independence of three nonparallel vectors (U2-LO4)

Use Gaussian elimination to determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

are linearly independent. If they are dependent, give a nontrivial linear combination of them that equals $\mathbf{0}$.

[Practice this activity in The Elimination Scratchpad](#)

Warning: Nonparallel vectors can still be dependent

In Testing independence of three nonparallel vectors (U2-LO4), the third vector is not parallel to either of the first two: its second entry is 1, while every scalar multiple of $\mathbf{v}_1$ has second entry 0; its first entry is 1, while

every scalar multiple of $\mathbf{v}_2$ has first entry 0. The first two vectors are also not parallel. Nevertheless, the three vectors are dependent because $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$. For three or more vectors, checking that no pair is parallel is not enough to establish linear independence: a vector can be a linear combination of the others without being a scalar multiple of any one of them.

The preceding example and activity demonstrate an algorithm to verify whether a given set of vectors is linearly independent.

Algorithm: A Test of Independence

To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

2. Solve the resulting **homogeneous** linear system.

3. If there is a **nontrivial** solution to the equation, then the

vectors are **linearly dependent**. Otherwise, the vectors are

linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M . Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem

If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span, $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, has a **unique** representation as a linear combination of the $\mathbf{x}_i$.

Why is this true?

Let $\mathbf{x}$ be an element of $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \dots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \dots + b_k\mathbf{x}_k,$$

then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \dots + (a_k - b_k)\mathbf{x}_k = 0.$$

Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = 0.$$

But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity: Testing independence in four dimensions (U2-LO4)

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

Basis and dimension

In geometry, we often describe lines as being ‘1-dimensional’, and planes as being ‘2-dimensional’. We can describe the notion of dimension more

precisely using linear algebra.

Theorem: A Fundamental Theorem

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity: Four vectors in three dimensions (U2-L04)

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

Definition: Bases of Vector Spaces

If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ in U is called

a **basis** of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.

Theorem: The Invariance Theorem

If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

The Invariance Theorem says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition: Dimensions of Vector Spaces

The **dimension** of a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U . We often write $\dim(U)$ for the dimension of U , and if $\dim(U) = d$, we say U is d -dimensional. By convention, the dimension of the trivial subspace $\{0\}$ is *zero*.

- **Term:** spanning set
Meaning: enough directions to generate the whole subspace
- **Term:** independent set
Meaning: no direction is redundant
- **Term:** basis
Meaning: just enough directions: spanning and independent
- **Term:** dimension
Meaning: number of directions in any basis

Activity: Dimensions of coordinate spaces (U2-LO4)

What is the dimension of $\mathbb{R}^3$?

Activity: Dimensions of coordinate spaces (U2-L04)

What is the dimension of $\mathbb{R}^4$?

Activity: Dimensions of coordinate spaces (U2-L04)

What is the dimension of $\mathbb{R}^n$?

Activity: The dimension of a parametric subspace (U2-L04)

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

Fact

Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U .

- Any spanning set for U contains a subset of vectors which form a basis for U .

Theorem

Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then B is linearly independent if and only if B spans U .

Why is this true?

Suppose B is linearly independent and $|B| = m = \dim(U)$. By A Fundamental Theorem, one cannot add any new vectors to the set B and remain linearly independent. But the referenced item says that if B is not already a spanning set for U , one can add new vectors to B from U , and remain linearly independent. Thus B must be a basis.

Conversely, if B spans U , then by the referenced item we can choose a subset of B which is a basis for U . But any proper subset of B contains fewer than m vectors, and thus by The Invariance Theorem, cannot be a basis for U . Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem

Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity: Adding a vector to an independent pair (U2-LO4)

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Explain why $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$. See the figure and the figure.

Activity: Adding a vector to an independent pair (U2-LO4) (continued)

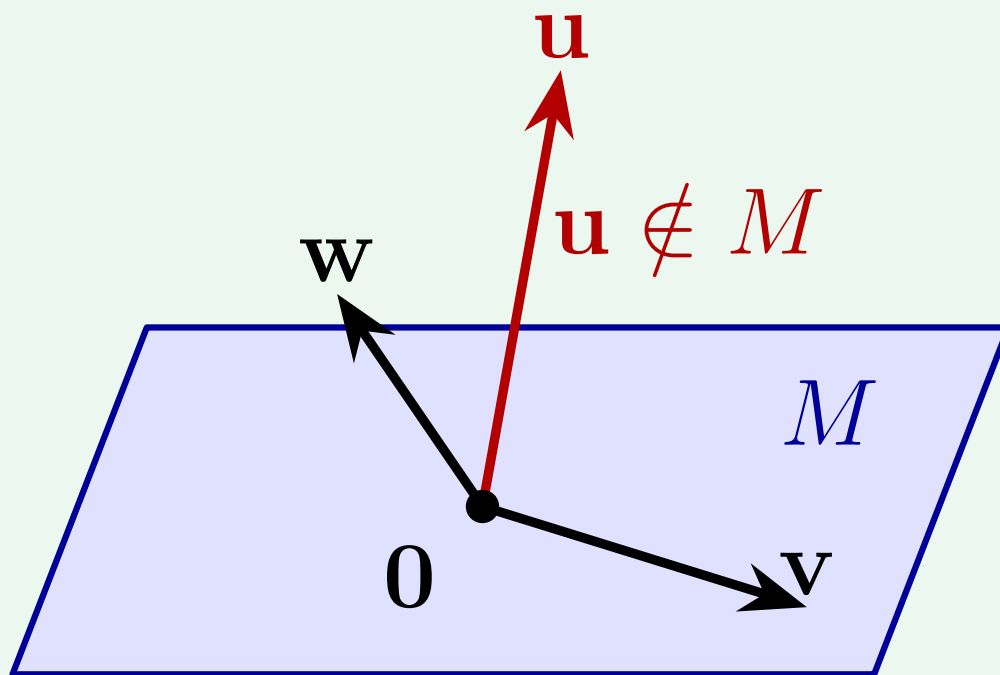


Figure. $\{u, v, w\}$ independent.

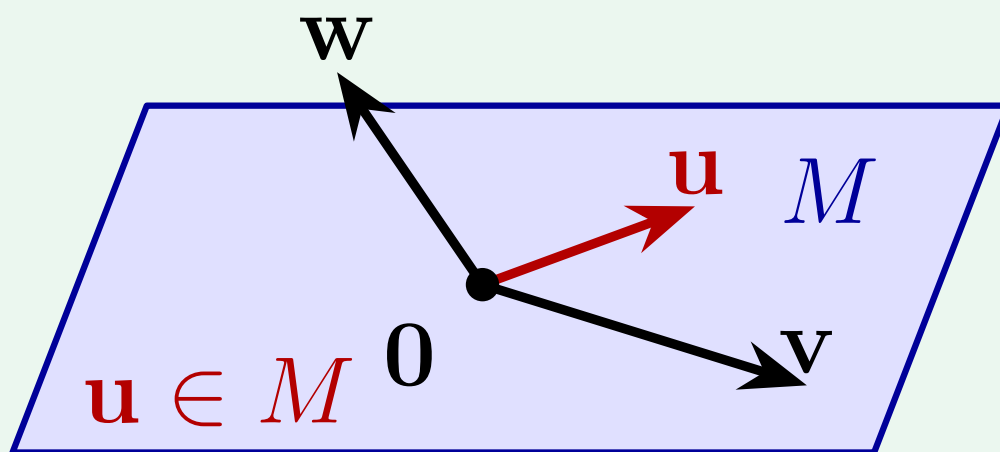


Figure. $\{u, v, w\}$ dependent.

Note

More generally, if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Activity: Extending a pair of vectors to a basis (U2-LO4)

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}.$$

Fact

The nonzero rows of a matrix in row echelon form are linearly independent.

Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination of a vector in the subspace are the coordinates.

Definition: Ordered Bases of a Subspace

Let $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the subspace U . We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an **ordered basis** of U .

As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are equal, and so describe the same basis of $\mathbb{R}^2$.

Given an ordered basis B for an m -dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its ‘coordinates’ with respect to the basis B , identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition: Coordinate Vectors With Respect to a Basis

Let U be an m -dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U . By the theorem, given any vector $\mathbf{v} \in U$, there exist unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1 \mathbf{b}_1 + \dots + t_m \mathbf{b}_m$$

and we define the **coordinate vector** of a vector $\mathbf{v} \in U$ with respect to B to be the m -vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}.$$

Note

If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U , then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector in $\mathbb{R}^m$.

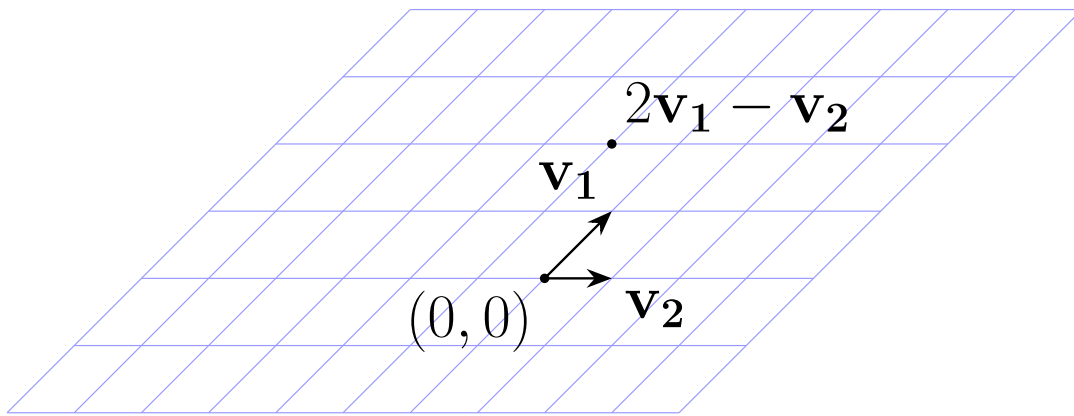


Figure. A slanted parallelogram grid determined by basis vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

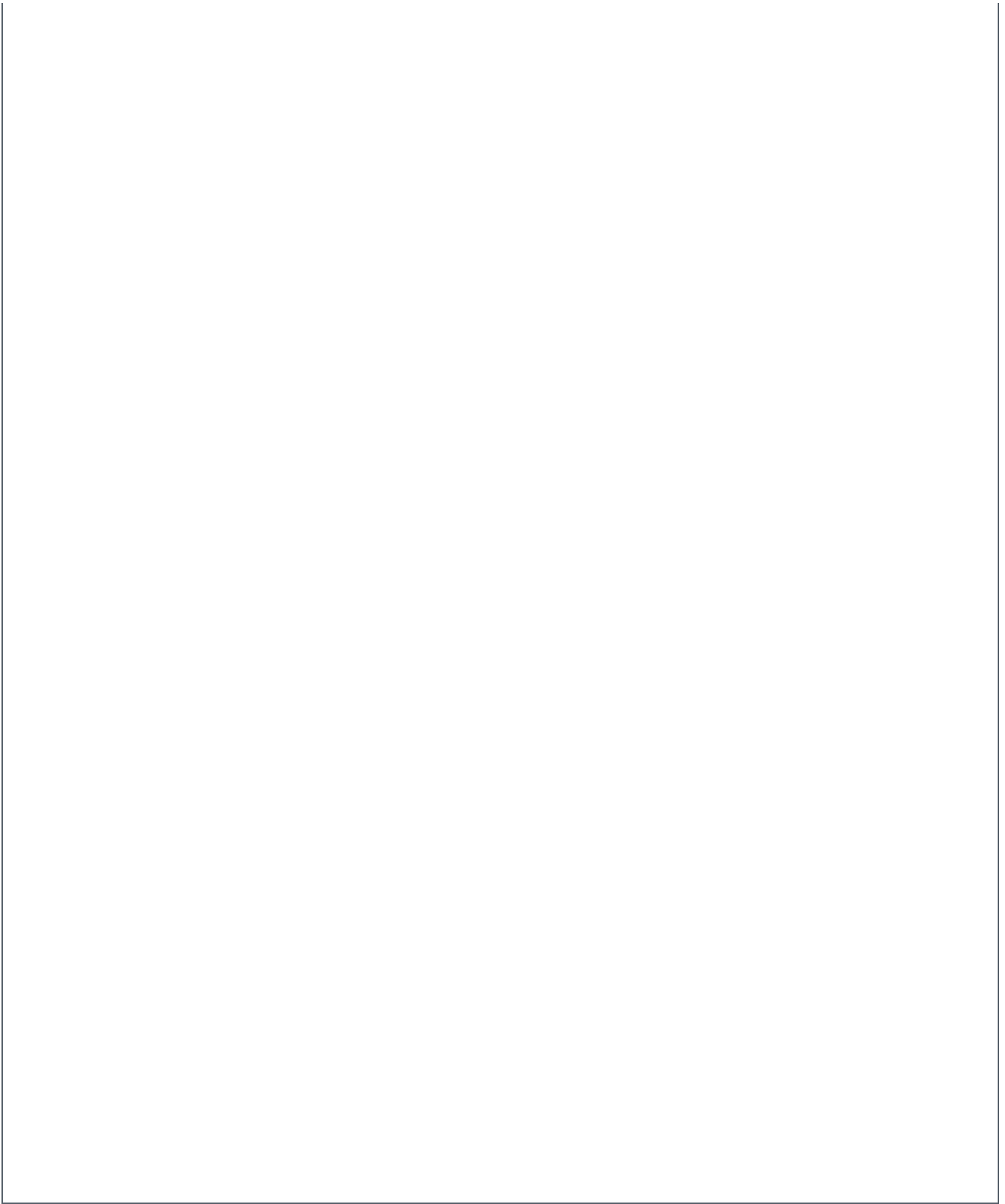
Activity: Coordinates relative to an ordered basis (U2-LO4)

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}.$$



Column and row space

In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined analogously as how they are defined for column vectors.

Definition: Column and Row Space

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$, span a *subspace* of $\mathbb{R}^m$ called the **column space** of A , denoted $\text{col}(A)$.
- The rows of A , considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the **row space** of A , denoted $\text{row}(A)$.

Note

The column space is the image space from Null Space and Image Space written with a column-based name:

$$\text{col}(A) = \text{im}(A).$$

The notation $\text{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\text{col}(A)$ emphasizes the columns that span those outputs.

Lemma

Let A and B be two row equivalent $m \times n$ matrices. Then

$$\text{row}(A) = \text{row}(B).$$

Why is this true?

Each time we apply a row operation to a matrix, we replace one or more of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A . But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A' . Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Lemma

If R is a matrix in row echelon form (recall Row echelon form and reduced row echelon form) then the non-zero rows of R are a basis of $\text{row}(R)$.

Why is this true?

The nonzero rows of R are linearly independent by the referenced item, and span $\text{row}(R)$ by definition. They thus by definition form a basis for $\text{row}(R)$.

Note

If A is row equivalent to a row echelon matrix R , then $\text{row}(A) = \text{row}(R)$. By the referenced item, the nonzero rows of R form a basis for this space. Thus $\dim(\text{row}(A))$ equals the number of nonzero rows, or equivalently the number of pivots, in R . In particular, this pivot count is independent of the choice of row echelon form.

Activity: Finding a basis for a span (U2-LO4)

Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Theorem: Rank Theorem

Let A denote any $m \times n$ matrix. Then

$$\dim(\text{col}(A)) = \dim(\text{row}(A)).$$

Write r for this common dimension. If A is row-equivalent to a matrix R

in row echelon form, then

- The r leading rows of R are a basis for $\text{row}(A)$.
- If the pivots of R lie in columns

$$j_1, j_2, \dots, j_r,$$

then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\text{col}(A)$.

Warning: Pivot columns come from the original matrix

Row reduction identifies pivot positions, but row operations do not preserve the original columns of A . When finding a basis for $\text{col}(A)$, use the pivot columns of the original matrix A , not the columns of the row-reduced matrix.

Activity: Selecting original pivot columns in SymPy (U2-LO3, U2-LO4)

The pivot-column rule translates directly into code. The assignment `R, pivots = A.rref()` stores the reduced matrix and its zero-based pivot indices separately. Rows and columns are indexed starting at zero: `A[i, j]` selects one entry, so `A[0, 2]` is the entry in the first row and third column.

A colon means “take all entries in this direction”: `A[:, j]` selects every row of column j , while `A[j, :]` selects every column of row j .

```
import sympy as sp

A = sp.Matrix([
    [2, 4, 0],
    [1, 2, 1],
    [3, 6, 2],
])
R, pivots = A.rref()
R, pivots
```

Activity: Selecting original pivot columns in SymPy (U2-LO3, U2-LO4) (continued)

Output:

```
(Matrix([
[1, 2, 0],
[0, 0, 1],
[0, 0, 0]]), (0, 2))
```

A **list comprehension** builds a list by evaluating an expression for each item in a sequence. Read `[A[:, j] for j in pivots]` as “for each pivot index j , take column j of A and put it in the list.”

```
columns = [A[:, j] for j in pivots]
columns
```

This is equivalent to the following loop; `append` adds one item to the end of a list.

```
columns = []
for j in pivots:
    columns.append(A[:, j])
```

1. What is `A[0, 2]`?
2. Predict the two column vectors in `columns`. Which mathematical columns do they come from? Why do they form a basis for $\text{col}(A)$?
3. What would `[R[:, j] for j in pivots]` return? Would these vectors form a basis for $\text{col}(A)$? Explain.
4. What would `[A[j, :] for j in pivots]` select? What goes wrong in `[A[:, j+1] for j in pivots]`?

Definition: Rank of a matrix

The rank of a matrix A , denoted $\text{rank}(A)$, is the common dimension of its column and row spaces:

$$\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A)).$$

- **Goal:** basis for $\text{row}(A)$
Row reduction tells you: nonzero rows of REF/RREF
Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\text{col}(A)$
Row reduction tells you: pivot column positions
Use these vectors: corresponding columns of original A
- **Goal:** basis for $\text{null}(A)$
Row reduction tells you: free variables
Use these vectors: basic solution vectors in $\mathbb{R}^n$

- **Goal:** rank
Row reduction tells you: number of pivots
Use these vectors: a number
- **Goal:** nullity
Row reduction tells you: number of free variables
Use these vectors: a number

Fact

It follows from Rank Theorem that for any matrix A , $\text{rank}(A) = \text{rank}(A^T)$. Indeed, the columns of A^T are the rows of A written as column vectors, so their spans have the same dimension. Hence

$$\text{rank}(A^T) = \dim(\text{col}(A^T)) = \dim(\text{row}(A)) = \text{rank}(A).$$

Theorem: Computing rank by row reduction

The rank of a matrix M is the number of pivots in any row echelon form row equivalent to M .

Why is this true?

By the referenced item, the pivot count equals $\dim(\text{row}(M))$, which is $\text{rank}(M)$ by Rank of a matrix.

This gives an algorithm for computing rank: use The Gaussian Elimination Algorithm to row-reduce the matrix to REF, then count the pivots.

Activity: Finding the rank of a matrix (U2-LO3)

Find the rank of the matrix

$$C = \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{bmatrix}$$

Activity: Finding the rank of a matrix (continued) (U2-LO3)

Find the rank of the matrix

$$D = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{bmatrix}$$



Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)
(continued)

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

What is $\dim(\text{row}(A))$?

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

Find a basis for the column space of A .

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

What is $\dim(\text{col}(A))$?

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

Find a basis for the nullspace of A .

[Practice this null-space calculation in The Elimination Scratchpad](#)

Definition

The **nullity** of a matrix A is equal to $\dim(\text{null}(A))$. It is denoted $\text{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The next theorem relates these two dimensions. But, first, let's do two examples. We can now state one of the preview patterns from Section 2.1, introduced in Two inputs, one output (U2-LO1, U2-LO6), in the language of null spaces.

Activity: Same output from a null-space direction (U2-LO3, U2-LO6)

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness of outputs?

Activity: Revisiting the height map with rank and nullity (U2-LO3)

Return again to A map that forgets height (U2-LO1, U2-LO3). Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P .

Remark

In Revisiting the height map with rank and nullity (U2-LO3), we found $\text{rank}(P) = 2$ and $\text{nullity}(P) = 1$, so

$$\text{rank}(P) + \text{nullity}(P) = 2 + 1 = 3.$$

The two transmitted directions and the one forgotten height direction account for all three input directions. The next theorem shows that this dimension count holds for every matrix.

Theorem: Rank-Nullity Theorem

Let A denote an $m \times n$ matrix of rank r . Then

- The $n - r$ basic solutions to the equation $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis for $\text{null}(A)$, so $\dim(\text{null}(A)) = n - r$.
- $\dim(\text{im}(A)) = r$.
- $\dim(\text{im}(A)) + \dim(\text{null}(A)) = n$.
- $\text{rank}(A) + \text{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \text{rank}(A) + \text{nullity}(A)$$

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4)

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{im}(B)$.

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4)

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{null}(B)$.

[Practice computing this RREF in The Elimination Scratchpad](#)

Activity: Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6)

The command `A.nullspace()` returns a Python list of basis vectors, each stored as a SymPy column matrix. With a nonempty list called `basis`, `basis[0]` selects its first vector, since list indices start at zero. The number of vectors is `len(basis)`, which gives the nullity. If the null space is $\{0\}$, the returned list is empty and has no first item.

```
import sympy as sp

A = sp.Matrix([
    [1, 0, 2],
    [0, 1, -1],
    [2, 1, 3],
])
basis = A.nullspace()
basis
```

Output:

```
[Matrix([
[-2],
[ 1],
[ 1]])]
```

With two numbers, `*` is ordinary multiplication; with a number and a matrix, it scales every entry. With two SymPy matrices, `*` means matrix multiplication, so `A * z` is the matrix-vector product when `z` is a column matrix. A flat list passed to `sp.Matrix`, such as `sp.Matrix([-2, 1, 1])`, creates that column matrix. This differs from NumPy arrays, where `A @ z` is the matrix-vector product and `*` acts entry by entry.

The command `sp.ones(3, 1)` creates a 3×1 column of ones. The attribute `z.T` is the transpose, a 1×3 row, and `z.shape` reports the pair (rows, columns).

Activity: Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6) (continued)

```
z = basis[0]
A * z
```

1. What are `len(basis)` and `z.shape`? Predict `A * z`.
2. Why must we select `basis[0]` before multiplying, rather than write `A * basis`?
3. Compute `A * sp.ones(3, 1)` and `z.T * A`. Explain why neither is the requested zero column.
4. What does the zero product tell you about $A(\mathbf{x} + t\mathbf{z})$?

Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.

Example: Redundant features in house data

Rows below represent houses. The three features are

first-level area, second-level area, total area,
measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X . The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features (U2-L06)

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Activity: Redundant square-footage features (U2-LO6) (continued)

1. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Compute $X\mathbf{c}$ and $X\mathbf{c}'$.

2. The two coefficient vectors tell different stories. Why do they make the same predictions?
3. What warning does this give about asking which feature is most important?

Further feature redundancy examples include:

- minutes and seconds, where $\text{seconds} = 60 \cdot \text{minutes}$;
- height in inches and height in centimeters, where $\text{centimeters} = 2.54 \cdot \text{inches}$;
- subtotal, tax, tip, and total bill, where $\text{total} = \text{subtotal} + \text{tax} + \text{tip}$;
- exam parts and exam total, where $\text{total} = \text{part 1} + \text{part 2} + \text{part 3}$.

Warning

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Auditing redundant features in code (U2-LO3, U2-LO6)

The matrix below stores the house features from Redundant square-footage features (U2-LO6).

```
import numpy as np

X = np.array([
    [9., 7., 16.],
    [11., 9., 20.],
    [14., 0., 14.],
    [8., 8., 16.],
])

z = np.array([1., 1., -1.])
c = np.array([3., 1., 0.])
c_alt = np.array([2., 0., 1.])

rank = np.linalg.matrix_rank(X)

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector c' .

Output:

```
(array([0., 0., 0., 0.]),
 2,
 array([34., 42., 42., 32.]),
 array([34., 42., 42., 32.]))
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?
3. Why do Xc and Xc' agree?
4. What does this say about interpreting coefficients when features are redundant?

Activity: A difference matrix forgets level (U2-LO3)

Recall the difference matrix from A difference matrix (U1-LO4, U1-LO5). For four input coordinates, let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

Activity: Reading the difference matrix in code (U2-LO3, U2-LO6)

```
D = np.array([
    [-1., 1., 0., 0.],
    [0., -1., 1., 0.],
    [0., 0., -1., 1.],
])

x = np.array([2., 5., 9., 10.])
ones = np.ones(4)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
(array([3., 4., 1.]),
```

Activity: Reading the difference matrix in code (U2-LO3, U2-LO6) (continued)

```
array([3., 4., 1.]),  
array([0., 0., 0.]))
```

1. What does $D @ x$ measure?
2. Why do the first two outputs agree?
3. What does $D @ \text{ones}$ say about the map?

Redundant features create null-space directions. If $Xz = 0$, then

$$X(c + tz) = Xc$$

for every scalar t . Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.